A Counterfactual Multi-Agent Framework for Adaptive Supplier Negotiation and Customer-Aware Supply Chain Switching

Department of Artificial Intelligence And Data Science

Project Phase -I

Project Review - II

ADI1411

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Problem

- Existing supplier negotiation mainly focuses on price, quality, delivery and risk.
- Future consequences of accepting an agreement are often not evaluated.
- Enterprises need to decide whether to retain, negotiate or switch suppliers.

Proposed Solution

- Multi-Agent AI for autonomous supplier negotiation.
- Counterfactual Negotiation Twin (CNT) predicts future supply-chain impact.
- Adaptive decision module selects Retain / Negotiate / Switch.
- Customer Impact module handles transparency after supplier switching.



Finalized System Design

Supplier Data	Collect supplier details like price, quality, delivery time and risk.
Supplier Evaluation	Score and rank suppliers based on important factors.
Multi-Agent Negotiation	Buyer and supplier agents exchange offers and negotiate automatically.
Counterfactual Negotiation Twin	Predict the future impact of each possible agreement (what-if analysis).
Supplier Comparison	Compare suppliers using the predicted results.
Retain / Negotiate / Switch	Decide whether to retain, negotiate further, or switch to a new supplier.
Customer Impact Analysis	Analyze how the supplier decision affects customers.
Customer Notification	Send notification to affected customers if there is a change.
Predicted vs Actual Performance	Compare predicted results with actual outcomes and improve the model.



Implementation Approach

Technologies

Python

Core development

Pandas

Data processing

Scikit-learn

Prediction and evaluation

MySQL / PostgreSQL

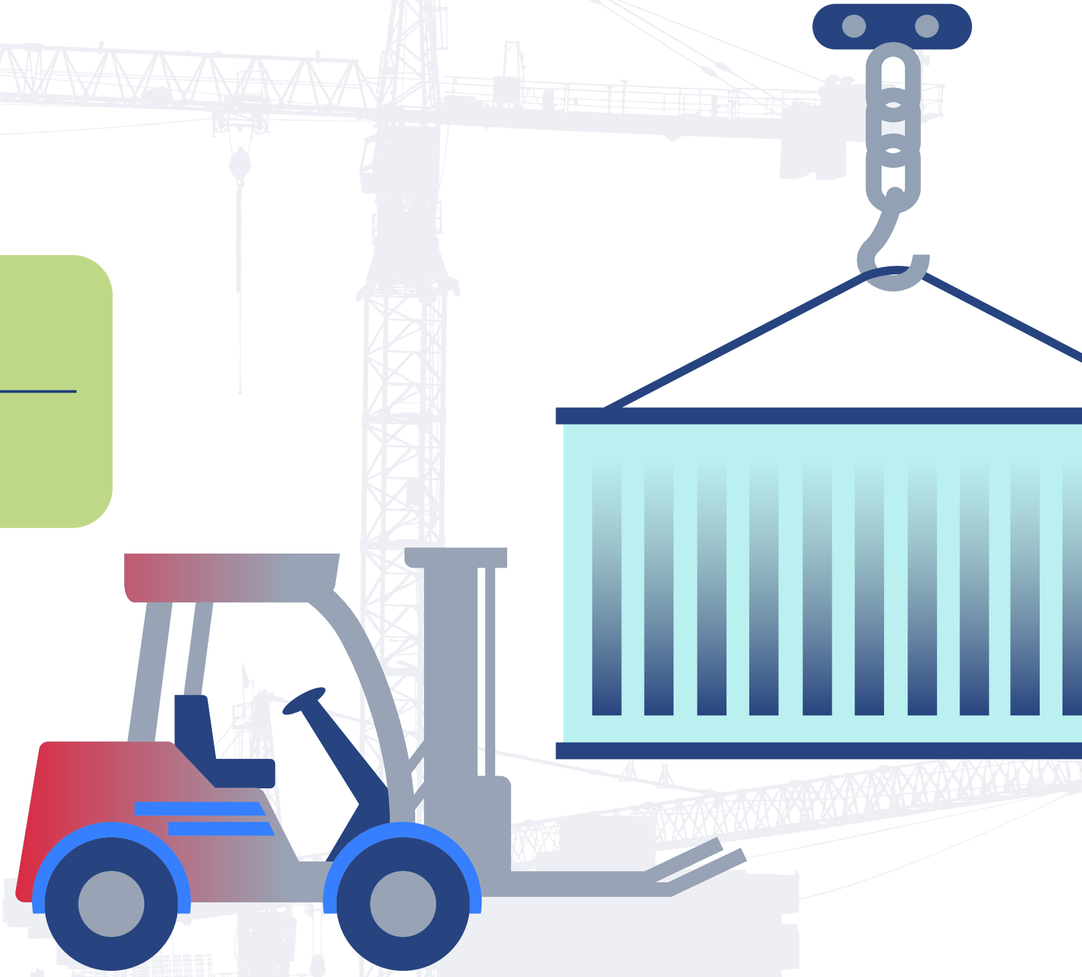
Data storage

LLM / Multi-Agent Framework

Warehousing & transportation

FastAPI / Flask

Reverse logistics & refunds



Development Approach

Data → Supplier Evaluation → Negotiation → Counterfactual Simulation → Decision → Customer Impact

Implemented Modules / Components

1.Supplier Evaluation Module

Evaluates suppliers using price, quality, delivery, reliability and risk.

2. Multi-Agent Negotiation Module

Enables Buyer and Supplier Agents to exchange offers and counteroffers.

3. Supplier Comparison Module

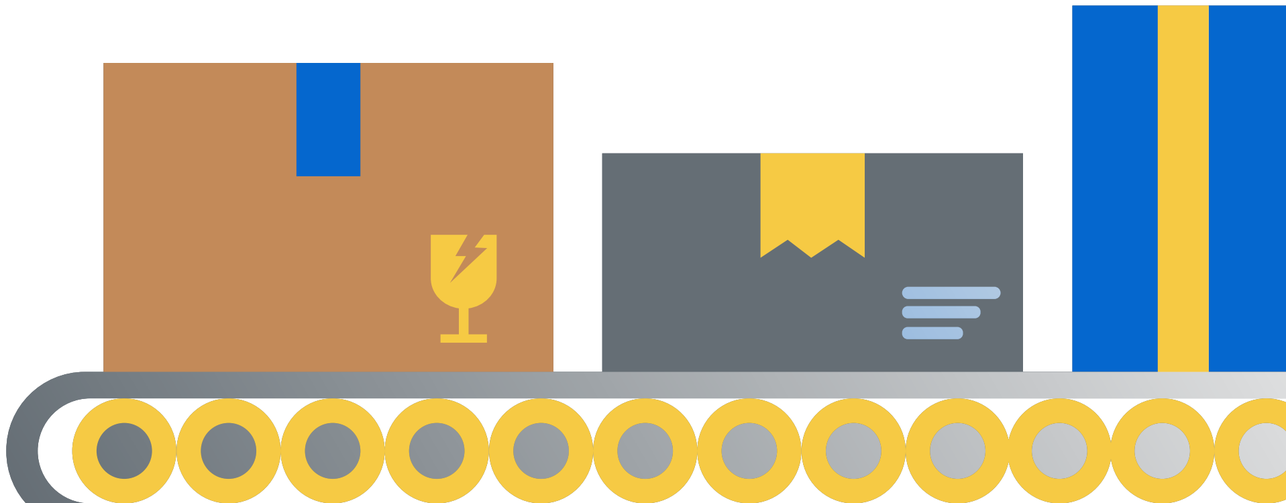
Compares candidate suppliers using multiple performance parameters.

4. Adaptive Decision Module

Determines Retain / Negotiate / Switch.

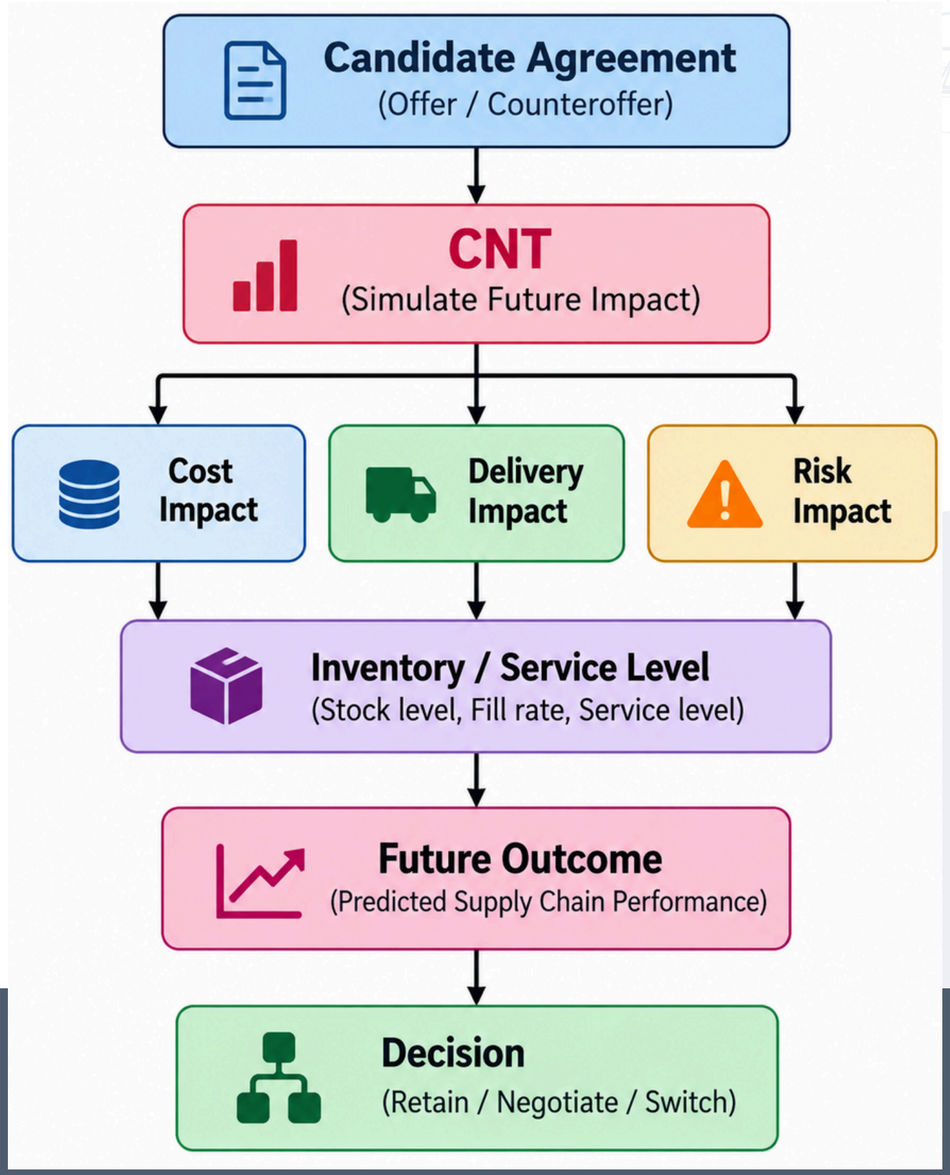
5. Counterfactual Negotiation Twin

Simulates possible outcomes of candidate agreements.

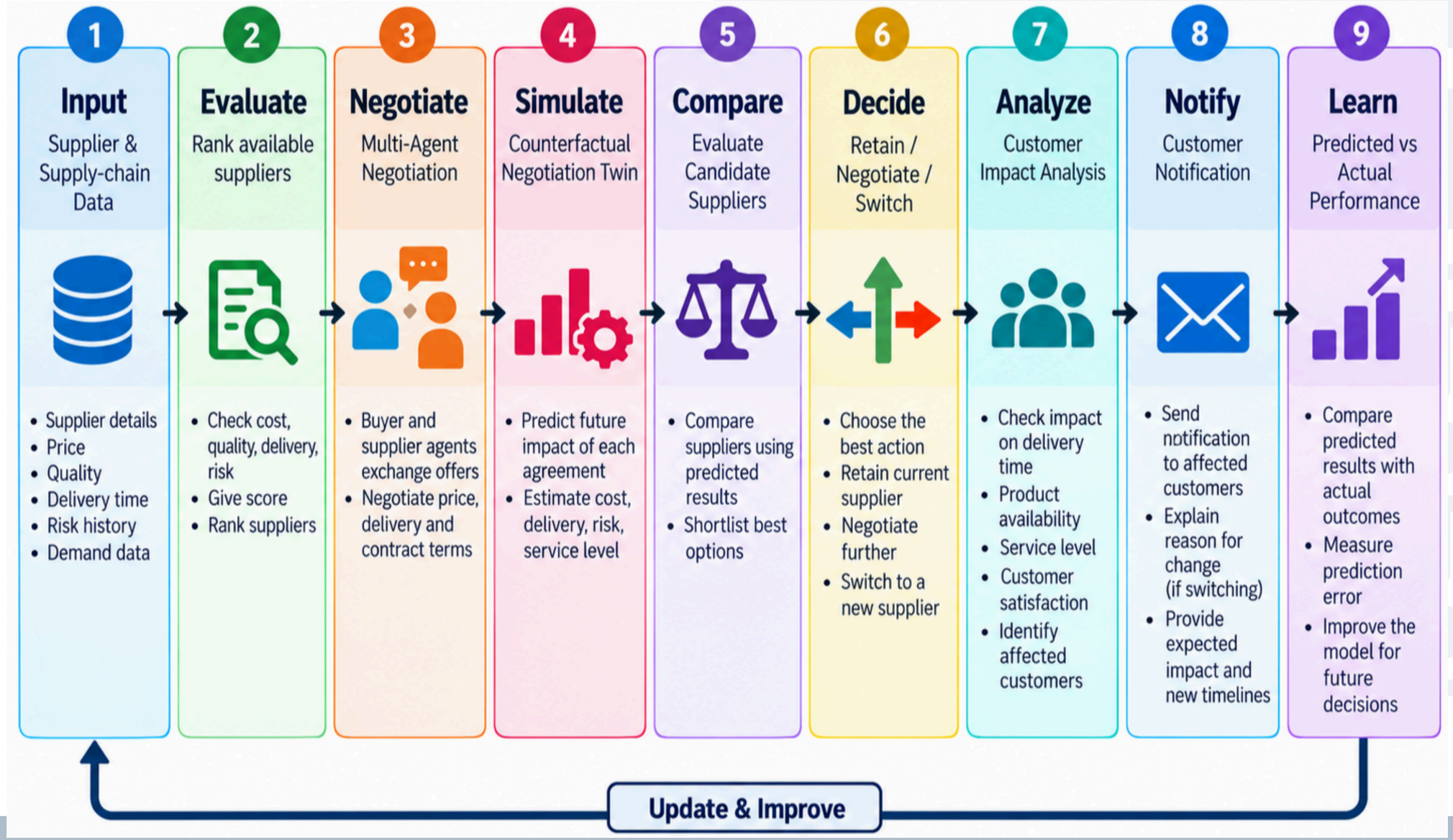


Core Implementation



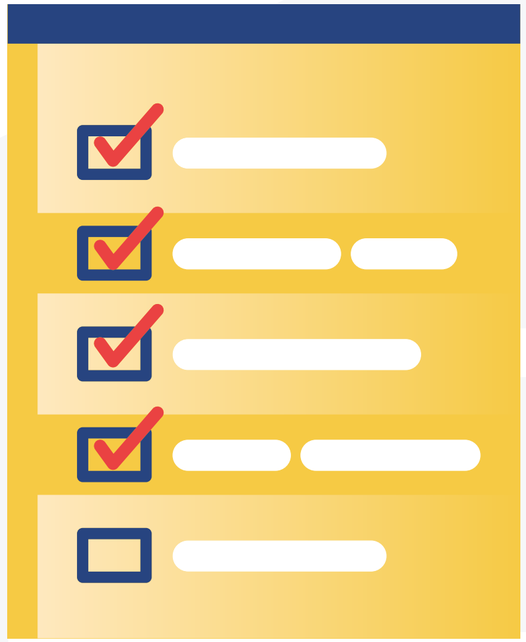


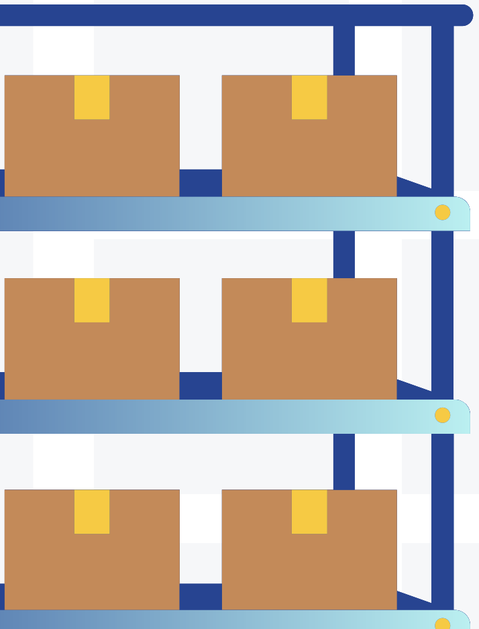
Demand Forecasting











Inventory Management

Balancing Supply & Demand

Just-In-Time (JIT) reduces inventory costs by receiving goods only as needed. Safety stock prevents stockouts from unpredictable demand or supply issues. Economic Order Quantity (EOQ) finds the ideal order size to minimize costs, and inventory turnover rate measures how often stock is sold and replaced, indicating management efficiency.

Goal: Minimize cost while maintaining availability.

Procurement & Sourcing

Strategic Supplier Management

The process involves evaluating vendors to ensure they meet the necessary standards, negotiating costs to achieve favorable terms, managing contracts effectively, and assessing risks to mitigate potential issues.

Strong supplier relationships improve reliability.



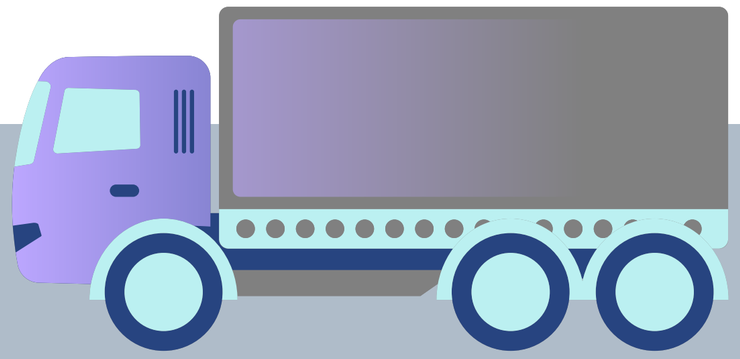
Current Progress & Limitations

Completed

- Problem and research gap identified
- System architecture finalized
- Multi-agent workflow designed
- Supplier evaluation module designed/implemented
- Counterfactual Negotiation Twin designed
- Supplier comparison and adaptive decision logic developed
- Customer-impact workflow designed

Current Limitations

- Limited real-world supplier datasets
- Counterfactual predictions depend on data quality
- LLM/API computational cost
- Real-world supplier integration is not yet available





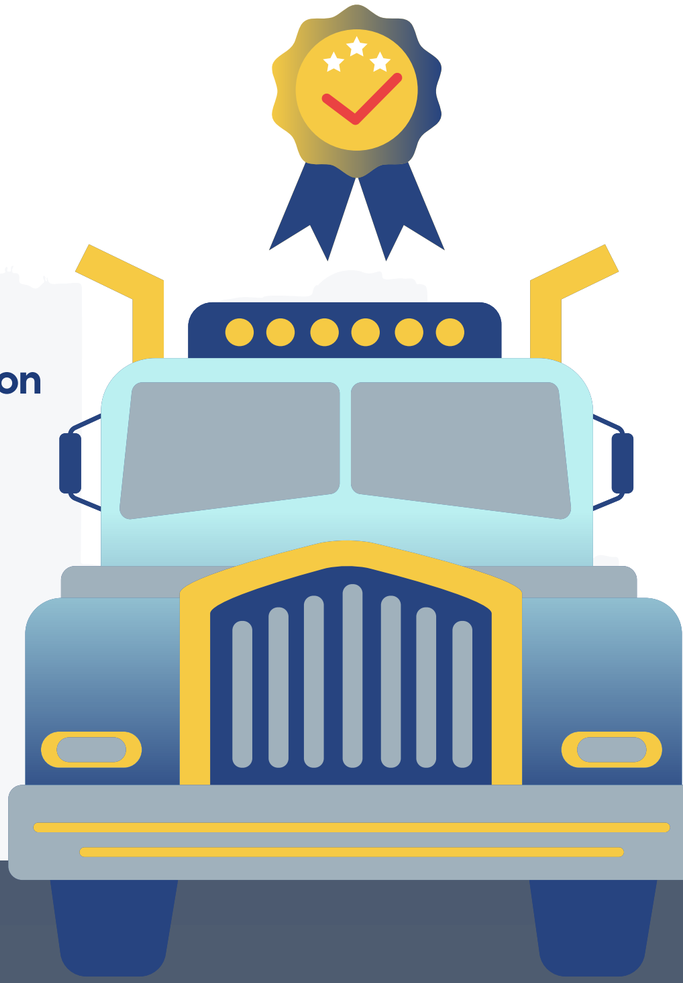
Phase-I Completion & Phase-II Plan

Phase-I – Completed

- Literature survey and research gap
- System architecture
- Agent design
- Supplier evaluation
- Negotiation workflow
- CNT design
- Adaptive supplier decision framework
- Initial prototype

Phase-II – Planned

1. Complete CNT implementation
2. Large-scale experiments
3. Compare with baseline methods
4. Validate prediction accuracy
5. Customer-impact testing
6. Dashboard/API integration
7. Final performance analysis
8. Research paper and final documentation





THANKYOU

